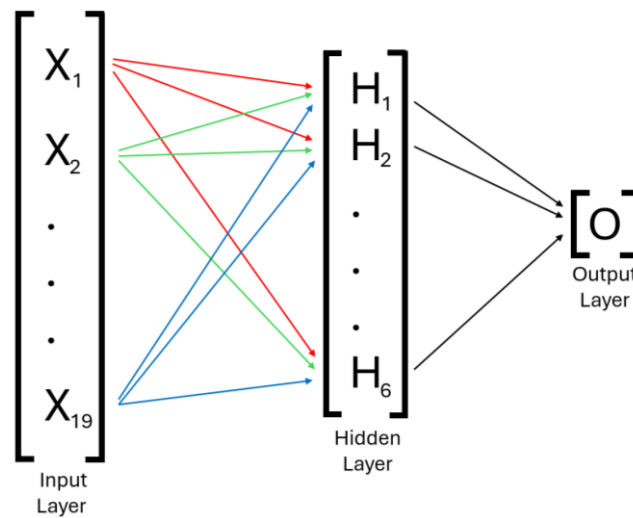


Lab 1 Report

Structure of Artificial Neural Network

The ANN is built up of three layers, with the first layer being the input layer of 19 nodes (neurons), followed by a hidden layer of 6 nodes and output layer of 1 node. To achieve forward feed, the input layer consists of nodes which are connected, each node representing an attribute of the data set. The hidden layer is connected to each node of the input layer with the sigmoid activation function. Output layer does binary classification using the sigmoid activation function.



Learning method to update weights of the network

The network uses the learning method of backpropagation algorithm, which updates the weights during training of the model to reduce the loss (difference from target value). The program first calculates the error term of the output layer using the formula:

$$E_o = o(1 - o)(t - o)$$

Where t is the target value, o is the predicted output, and E_o is the error term.

It can then calculate the error term of the hidden layer using the formula:

$$E_h = (1 - o_h) \cdot o_h \cdot \sum(w_o \cdot E_o)$$

Where o_h is the output of the hidden node, w_h is the weight from the node to the output layer, and E_h is the error term of the hidden node.

And updating weights using the formula:

$$w_o = w_o + \eta \cdot E_o \cdot o_h$$

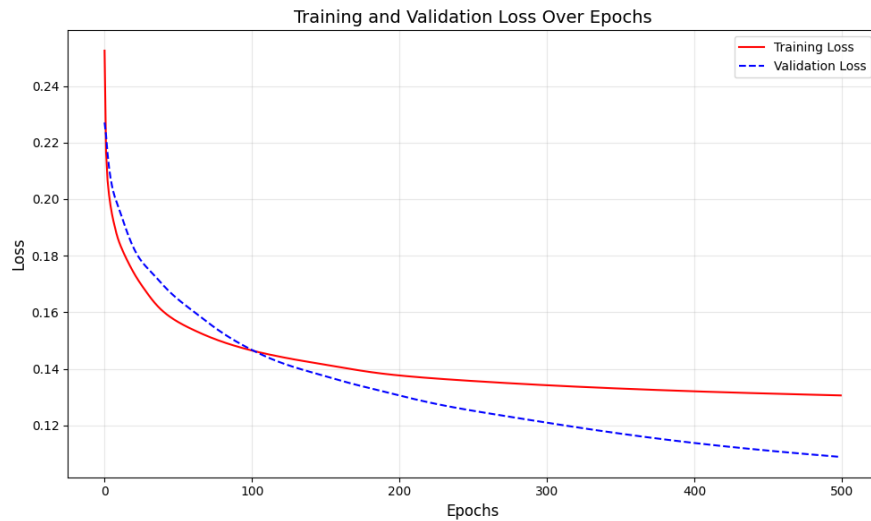
$$w_h = w_h + \eta \cdot E_h \cdot x_i$$

Where w_h is a weight from a node from the input layer to a node in the hidden layer, x_i is an input value, and η is the learning rate.

Learning rate of 0.1 is chosen to allow for small enough steps during gradient search, to avoid too large updates after each iteration.

Figure

Change of validation accuracy during learning process by the ANN. Only decreasing error due to limited number of epochs, otherwise an increase of validation error would suggest that the training is finished.



Classification accuracy of ANN on the test data set

When testing the finished model, it showed an accuracy of 86.5029%.